

ViroReportR: An R package for automated short-term forecasting and report generation of viral respiratory infections

Rebeca Cardim Falcao^{1,2}, Nelson Tang¹, Nirupama Tamvada¹, Cesar Wong¹, and Michael A. Irvine^{1,3,4}

¹ BC Centre for Disease Control, Canada ² Department of Mathematics, University of British Columbia, Canada ³ Faculty of Health Sciences, Simon Fraser University, Canada ⁴ Department of Statistics, University of British Columbia, Canada  Corresponding author

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Software

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Summary

Respiratory viruses can circulate year-round with some showing strong seasonal patterns. Consistent and reliable forecasts are essential for public health to provide information for healthcare planning, interventions such as vaccination, and public communication about population level risks. ViroReportR is an R package that automates short-term forecasts and end-to-end routine report generation for common respiratory viruses (e.g. Sars-CoV-2, Influenza A and Respiratory Syncytial Virus) from routine surveillance data. Its implementation lies on using a renewal equation approach to estimate the instantaneous time-varying reproduction number (R_t) via the EpiEstim package (Cori et al., 2013), and provides projections at specific time horizons defined by the user with validation metrics in a standardized HTML format. The tool is formulated for operational use by surveillance teams and public health officials, whose needs comprise robust, transparent, and reproducible results with minimal configuration. The package is available on CRAN and Github with documentation, examples, and vignettes.

Statement of need

Respiratory viruses cause on-going burden across health systems, especially during respiratory season where their circulation within a population is more prevalent. Short-term forecasts of cases related to respiratory infection are key epidemiological indicators to support public health decision-makers. Routine and reproducible multi-pathogen forecast reports should contain standardized uncertainty, model validation, and a reproducible workflow to aide the public health community, researchers, and ensure public trust. ViroReportR addresses this issue directly by providing an automated reporting workflow, where users supply daily counts in a three-column dataset (date, disease_type, confirm), optionally specifying serial interval parameters, forecast horizon and seasonal patterns, and obtain a HTML report with forecasts plots, uncertainty intervals, (R_t) estimates, and validation metrics across pathogens. ViroReportR focuses on operational transparency and replicability, making it a well-suited and reliable tool to public health surveillance teams and policy makers who need daily to weekly briefs generated by routine data.

State of the field

There are a variety of packages that provides tools for surveillance, epidemic parameter estimation, nowcasting, or time-series forecasting. However, they usually target a different

user group or different layers of the analytic stack compared to ViroReportR. None provide a fully automated multi-pathogen reporting tool for routine public health surveillance. Below, we briefly summarize the most relevant packages and provide clarity on how ViroReportR is distinctively situated and filling a specific gap.

1. EpiNow and EpiNow2: both of these packages estimate time-varying reproduction number, growth rate, and doubling time using a range of open-source tools. EpiNow is now deprecated in favor of EpiNow2. The aim of these packages is to provide a framework that is informed by community feedback and actively supported, avoiding potential limitations of naive implementations. Forecasting is also available, by extending the approach used for estimation. However, outputs are provided in the form of R objects or plots and not as a combined report suitable for public health. Due to the Bayesian modelling framework utilized in these packages, computational time may be a constraint for users especially for longer time-series.
2. surveillance: this package focuses on statistical methods for monitoring and modelling of time series of counts, proportion and categorical data, common in epidemiological settings. With a specific focus on aberration detection in count time series of communicable diseases, outbreak detection, and endemic-epidemic modelling framework, including visualization and simulation tools. Similar to EpiNow and EpiNow2 this package doesn't offer short-term forecasts reporting tools.
3. Sitrep: provides report templates for epidemiological surveys and outbreaks reports. The templates are intended for *Medecins Sans Frontieres* analyses, however, have a generic format that can be used by general epidemiological community. Their reports focuses on epidemiological indicators and standardize common epidemiological analyses, such as attack rates and descriptive analyses of outbreak data. It does not provide short-term forecasts of daily counts across pathogens.
4. epi_predict: offers a framework for building transformation and forecasting pipelines tailored for epidemiological time-series data. It uses a variety of forecasting techniques and also provides back testing tools for validation. However, it does not offer a tool for automated report generation, requiring the user to assemble their own reporting pipeline on top of epi_predict.
5. PROF: package designed for respiratory disease forecasting. PROF supports modelling of influenza and COVID-19 using either compartmental modelling or a baseline statistical model. However, it doesn't generalize for arbitrary user-defined pathogens and does not include a unified automated reporting structure for surveillance purposes.

Across CRAN there are a variety of packages providing strong modelling capabilities, reporting templates, forecasting and outbreak detection. However, none of them integrates multiple component of the epidemiological analysis stack into a single automated workflow. ViroReportR: 1. accept multi-pathogen surveillance data in a format of daily counts with columns date, confirm, and disease_type. 2. produces (R_t)-based forecasts with uncertainty quantification. 3. performs backtesting for validation and summarizes standard forecasting metrics. 4. renders a standardized HTML report for use on weekly briefs.

ViroReportR stands between low-level modelling libraries, such as EpiNow2, and EpiEstim, and reporting templates libraries such as Sitrep, by integrating modelling, forecasting and reporting into a single tool. Filling a unique niche, ViroReportR offers an operationally tailored multi-pathogen reporting tool specifically for virus respiratory pathogens.

Software design

Our key design goals were to provide a flexible and easy to use report generation tool, while keeping the overall structure of the report fixed. The ViroReportR package works around a main function `generate_forecast_report()` that calls all other functions for data processing, fitting, forecasting, validation, and renders a final markdown file based on the template provided in the package.

We designed the workflow to be as simple as possible, only needing a cleaned data file of the daily counts of a given set of respiratory viruses. The number of pathogen types used to generate the report are inferred directly from the structure of the data. The required format of the data is 3 columns with date, disease_type and confirm (confirmed cases), which is in keeping with the design in EpiNow2. If the user has added any additional viruses other than Influenza A, SARS-CoV2 and RSV, they will also need to add serial interval distribution parameters for those viruses. The overall text and explanations on the report are not dependent on results. Adding new virus provides more tabs, plots, and summary tables on the final html report. In addition, some respiratory viruses follow a clear respiratory season, whereas others may not. Thus, we include an option for users to provide seasonal period for each virus as needed. Currently, we provide estimations using the EpiEstim packages (Cori et al., 2013), but we have designed ViroReportR to be model agnostic and allow for inclusion of other model estimation methods in future releases.

Research impact statement

The ViroReportR package was developed primarily to streamline forecasting and reporting at the BC Centre for Disease Control. As such, it has been deployed for monitoring by the Public Health Response team during the 2025/26 respiratory season and hosted on the Centre's Posit Connect infrastructure. It has also been deployed to monitor and forecast the number of COVID-19, Influenza A, and Respiratory Syncytial Virus at mass gathering events within Vancouver, British Columbia during the 2026 FIFA world cup.

AI usage disclosure

AI tools were not used in the writing of this manuscript. AI coding assistance tools such as Copilot were used in the generation of tests and code documentation. All code was submitted as a pull request on Github and independently checked by another team member before merging to ensure its quality and validity.

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References

- Cori, A., Ferguson, N. M., Fraser, C., & Cauchemez, S. (2013). A New Framework and Software to Estimate Time-Varying Reproduction Numbers During Epidemics. *American Journal of Epidemiology*, 178(9), 1505–1512. <https://doi.org/10.1093/aje/kwt133>